CHENYE YANG

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EDUCATION

University of California, Davis - United States *Sep 2022 – Present*
Ph.D. Candidate in Electrical and Computer Engineering **Degree GPA: 3.87/4.0**

Columbia University - United States *Sep 2019 – Apr 2021*
M.S. Advanced Research Specialization in Electrical Engineering **GPA: 3.8/4.0**
Nikola Tesla Electrical Engineering Scholarship

Xi'an Jiaotong University - China *Sep 2015 – Jun 2019*
B.Eng. in Automation **Grade: 89.42/100**
Honors Electronic and Information Engineering Program (QianXuesen Class)

RESEARCH EXPERIENCE

Adversarially Robust DPO *Prof. Lifeng Lai, UC Davis, Jan 2026 – Present*
- Study adversarial robustness of DPO against preference poisoning. *(Manuscript under review)*

Preference Poisoning Attack on DPO *Prof. Lifeng Lai, UC Davis, May 2025 – Dec 2025*
- Studied targeted preference poisoning attacks against offline RLHF/DPO from a theoretical perspective.
- Showed that label flipping in log-linear DPO induces a parameter-independent gradient shift, reducing the attack design to a binary sparse approximation problem.
- Proposed two efficient attack algorithms: Binary-Aware Lattice Attack and Binary Matching Pursuit Attack.
- Proved theoretical guarantees for attack feasibility, binary enforcement, minimum-flip recovery, coherence-based support recovery, and impossibility of successful K-flip attacks.
- Experiments on synthetic data and the SHP dataset validate the theory. *(Accepted at ICML 2026)*

Preference Robustness of Online RLHF *Prof. Lifeng Lai, UC Davis, Dec 2023 – May 2025*
- Studied preference robustness of RLHF algorithms with online human feedback from a theoretical perspective.
- Investigated preference attacks that force online RLHF to learn a target suboptimal policy with small attack cost.
- Proposed an adversarial human-feedback poisoning strategy and proved its success in misleading online RLHF.
- Proposed a robust-defense online RLHF and proved its robustness to any attacker with bounded attack cost.
- Simulation results validate the theoretical analysis. *(Published on IEEE TSP 2025)*

Reward Robustness of Bandit *Prof. Lifeng Lai, UC Davis, Sep 2022 – Feb 2024*
- Studied reward attacks on stochastic multi-armed bandits in non-stationary environments.
- Investigated attackers who force the victim algorithm to mostly choose a suboptimal arm with small attack cost.
- Considered three general scenarios with different environments, algorithms, and attacker information.
- Proposed three attack strategies and proved their success in terms of target-arm selection and attack cost.
- Proposed a non-stationary bandit defense method and proved its robustness to any attacker with bounded attack cost.
- Simulation results validate the theoretical analysis. *(Published at ACSSC 2023 and on IEEE TSP 2024)*

Edge AI System for Smart Home *Prof. Xiaofan (Fred) Jiang, Columbia University, Jun 2021 – Sep 2021*
Summer Research Assistant in Columbia Intelligent and Connected Systems Lab.
- An auto-discover system able to allocate sensors and actuators according to the smart home applications user specified; deployed in multiple lab rooms. *(Published at SenSys 2021)*
- Designed the MCU/sensors-to-application architecture and built the hardware testbed from scratch.

Fever Screening System *Prof. Xiaofan (Fred) Jiang, Columbia University, Jan 2020 – May 2021*
Research Intern in Columbia Intelligent and Connected Systems Lab.
- A low-cost system based on RGB-thermal cameras for continuous fever screening of multiple people without human interaction; deployed in a restaurant and hospital clinic in New York City. Featured in Columbia News.
- Implemented multiple RGB-thermal heads matching, trained and deployed YOLOV3 (head detection) and FSA-Net (head orientation regression), estimated distance using non-identical RGB and thermal camera.
- *(Published at SenSys 2020, CPHS 2021, and IPSN 2022)*

Khameleon Scheduler in RL *Prof. Eugene Wu, Columbia University, Jul 2020 – Sep 2020*
Research Intern in WuLab Columbia University.
- A server-side scheduler involving a complex optimization based on available resources, predicted user interactions, and response quality levels to maximize user-perceived interactivity and satisfaction in real-time.
- Implemented Q-learning and SARSA-based prefetching schedulers in a simulated RL environment to trade off latency and response quality in cloud-based interactive applications.

Cyber-Physical Energy Systems *Prof. Jiang Wu, Xi'an Jiaotong University, Oct 2017 – Feb 2019*
Member of XJTU Information-technology Talent Program.
Research Intern in Ministry of Education Key Lab for Intelligent Networks and Network Security.
- Proposed centralized & distributed K-means, using weighted combination of features from PCA and prior knowledge & parameter consensus and feature transferring method. Analyzed urban power grid data. *(Published at CCC 2018)*

PUBLICATIONS

[1] **Chenye Yang**, Weiyu Xu, Lifeng Lai, "Adversarially Robust Direct Preference Optimization against Preference Label Flip Attacks" *Manuscript under review*.

[2] Mo Lyu, **Chenye Yang**, Lifeng Lai, "Efficient Reward Manipulation Attacks Against GRPO under RLVR" *Manuscript under review*.

[3] **Chenye Yang**, Weiyu Xu, Lifeng Lai, "Efficient Preference Poisoning Attack on Offline RLHF" *The 43rd International Conference on Machine Learning (ICML 2026)*.

[4] Mo Lyu, **Chenye Yang**, Guanlin Liu, Lifeng Lai, "Dueling Bandit: Adversarial Attack and Robust Defense" *Manuscript under review*.

[5] Mo Lyu, **Chenye Yang**, Guanlin Liu, Lifeng Lai, "Adversarial Post-Action Attacks on Dueling Bandits" *The 59th Asilomar Conference on Signals, Systems, and Computers (ACSSC 2025)*.

[6] **Chenye Yang**, Mo Lyu, Guanlin Liu, Lifeng Lai, "Human Feedback Attack on Online RLHF: Attack and Robust Defense" *IEEE Transactions on Signal Processing, Volume: 73 (IEEE TSP 2025)*.

[7] **Chenye Yang**, Guanlin Liu, Lifeng Lai, "Stochastic Bandits With Non-Stationary Rewards: Reward Attack and Defense" *IEEE Transactions on Signal Processing, Volume: 72 (IEEE TSP 2024)*.

[8] **Chenye Yang**, Guanlin Liu, Lifeng Lai, "Reward Attack on Stochastic Bandits with Non-stationary Rewards" *The 57th Asilomar Conference on Signals, Systems, and Computers (ACSSC 2023)*.

[9] Kaiyuan Hou, Yanchen Liu, Peter Wei, **Chenye Yang**, Hengjiu Kang, Stephen Xia, Teresa Spada, Andrew Rundle, Xiaofan Jiang, "A Low-Cost In-situ System for Continuous Multi-Person Fever Screening" *The 21st ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN 2022)*.

[10] Stephen Xia, Rishikanth Chandrasekaran, Yanchen Liu, **Chenye Yang**, Tajana Simunic Rosing, Xiaofan Jiang, "Demo Abstract: A Drone-based System for Intelligent and Autonomous Homes" *The 19th ACM Conference on Embedded Networked Sensor Systems (SenSys 2021), Best Demo Award*.

[11] Peter Wei, Yanchen Liu, Hengjiu Kang, **Chenye Yang**, Xiaofan Jiang, "A Low-Cost and Scalable Personalized Thermal Comfort Estimation System in Indoor Environments" *The 1st International Workshop on Cyber-Physical-Human System Design and Implementation (CPHS 2021)*.

[12] Peter Wei, **Chenye Yang**, Xiaofan Jiang, "Poster Abstract: Low-Cost Multi-Person Continuous Skin Temperature Sensing System for Fever Detection" *The 18th ACM Conference on Embedded Networked Sensor Systems (SenSys 2020)*.

[13] Pengyuan Liu, **Chenye Yang**, Jiang Wu, "Hybrid Features Based K-means Clustering Algorithm for Use in Electricity Customer Load Pattern Analysis" *The 37th Chinese Control Conference (CCC 2018)*.

PROJECT EXPERIENCE

Error Correcting Code *Prof. Ulrich Jetzek (from Fachhochschule Kiel), UC Davis, Apr 2023 – Jun 2023*
- Converted TXT/PNG/WAV files to bit streams; encoded with (7,4) Hamming and (n, k) systematic cyclic codes.
- Simulated binary symmetric channels; decoded with syndrome/error-trapping decoders and reconstructed files.

Large Scale Stream Processing *Prof. Deepak S. Turaga, Columbia University, Jan 2020 – May 2020*
- Built a multithreaded Python data generator and stream processor with a database-backed web control interface.
- Designed socket/HTTP data paths and WebSocket control channels across components.

Sparse Models for High-Dimensional Data *Prof. John Wright, Columbia University, Jan 2020 – May 2020*
- Built a system for fast moving-object detection in high-resolution video; tested on sports video and daily scenes.
- Used super-resolution reconstruction to speed processing while preserving high-resolution output.

Internet of Things *Prof. Xiaofan (Fred) Jiang, Columbia University, Sep 2019 – Dec 2019*
- Programmed ESP8266 for I/O, APIs, server mode, MongoDB connectivity, and gesture recognition application.
- Built a distributed scalable system to monitor soil conditions over long time at multiple locations via LoRa wireless communication; data stored in MongoDB; website designed for data visualization and systems control.

Statistical Learning *Prof. Predrag Jelenkovic, Columbia University, Sep 2019 – Dec 2019*
- Reproduced the results of identification/authentication in *Smartphone and Smartwatch-Based Biometrics Using Activities of Daily Living* with WISDM Dataset using decision tree, random forest and K-NN.
- Trained SVM on the same dataset and compare different methods.

Random Matrix Theory and Application *Prof. Ori Shental, Columbia University, Sep 2019 – Dec 2019*
- Reproduced the results of uniform samples in *Bandlimited Field Reconstruction for Wireless Sensor Networks*.
- Tested Gaussian sampling and analyzed the resulting differences.

TEACHING EXPERIENCE

Ph.D.	Reader	*EEC 161 Applied Probability*	Winter 2026
	Teaching Assistant	*EEC 256 Reinforcement Learning*	Spring 2025
		ENG 006 Engineering Problem Solving	Winter 2025
		EEC 173A / ECS 152A Computer Networks	Winter 2024 & Winter 2023
M.S.	Teaching Assistant	*E4764 IoT Intelligent and Connected Systems*	Fall 2020

WORK EXPERIENCE

WiLO Networks Inc. ***Network Engineer Intern*** *Oct 2021 – Apr 2022*
Early-stage startup; owned end-to-end development of the smart-farm prototype across hardware, network, and cloud.
- Architected and implemented a low-cost LoRa mesh network on ESP32 + SX1276 (LoRa PHY), including auto-join/routing protocol and robustness strategies; validated via Duke Smart Farm field tests and NS-3 simulation.
- Integrated and modified MCCI LoRaWAN LMIC at the register level to control SX1276 mode transitions, interrupts, and Tx/Rx scheduling; aligned with ESP32 OS.Job timing for reliable communications.
- Delivered sensor-to-cloud data pipeline (sensors → MCU → LoRa → Raspberry Pi gateway → InfluxDB/Azure) enabling data fusion and ML-based monitoring.
- Standardized reusable MCU libraries + KiCad schematic/footprint libraries for team-wide reuse, and designed and assembled a backscatter-tag PCB in KiCad (component selection + layout).

TECHNICAL SKILLS

Programming:	Python, C/C++/SystemC, MATLAB, HTML, R, Verilog HDL
Software & Tools:	**AI Coding:** OpenAI Codex, Claude Code
	Platform: VS Code, PlatformIO
	Design: KiCad, Autodesk Inventor, Altium Designer, LabVIEW, FDTD Solutions
	Others: GitHub, Linux, LaTeX, Docker, PyTorch, OpenCV, Spark, AWS, GCP

AWARDS

Gold Reviewer Award	*The 43rd International Conference on Machine Learning (ICML)*	*2026*
Best Demo Award	*The 19th ACM Conference on Embedded Networked Sensor Systems (SenSys)*	*2021*
Nikola Tesla Scholarship	*Columbia University*	*2019*
University Third Scholarship	*Xi'an Jiaotong University*	*2018*
Siyuan Scholarship	*Xi'an Jiaotong University*	*2017*
Outstanding Student	*Xi'an Jiaotong University*	*2017*
Provincial First Prize	*China Undergraduate Mathematical Contest in Modeling*	*2016*
National Second Prize	*VEX Robotics China Open*	*2016*